

Team Control Number

**16471**

Problem Chosen

**A**

2026

HiMCM/MidMCM

Summary Sheet

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## An Evacuation Sweeping Strategy Based On Minimum Spanning Tree

In this paper, we present a thorough mathematical model for **the optimization of evacuation sweeps** in multi-story buildings during emergency conditions such as fires and gas leaks. Recognizing the critical need for efficient responder deployment in complex environments, we develop an approach that integrates graph theory with optimization algorithms to offer responder strategies and guide rapid, orderly evacuation.

Our model begins by using **Depth-First Search (DFS)** in simple single-floor layouts, a six-room office building with a long corridor connecting the two exits. We ensured complete clearance throughout the building in 31.2 seconds when deploying two responders. Then, for more complicated layouts, we partition a **Minimum Spanning Tree (MST)** using Kruskal's algorithm, assigning different zones to multiple responders. This approach balances workload among responders, ensuring a minimal time required for clearance.

When applied to **two real-world cases**, the model showcases both effectiveness and efficiency: the White House (43 rooms on 2 floors) requires **520.8 seconds** with two responders, while a five-story L-shaped school building needs **383.1 seconds** with the assistance of five responders. Our model incorporates realistic factors such as the effect of fire spread, smoke hindrance, and prioritized areas where conducting multiple sweeps is necessary.

Moreover, our sensitivity analysis reveals moderate sensitivity to visibility and search efficiency parameters, with 10% variations resulting in approximately 10% change in overall sweeping time on average. The MST partitioning strategy also **proves robust against parameter fluctuations**, as the optimal strategy it provides persists regardless of sweeping time variations. Compared to traditional approaches, our model also depends little on technological equipment such as sensors and cameras, though their presence would improve the evacuation process.

Overall, **our model ultimately provides a valuable approach** for emergency plannings across varying building structures and layouts. Its combination of MST partitioning and DFS guarantees a computationally efficient and practically optimal solution to evacuation processes, maximizing occupant safety.

**Keywords:** Sweeping Strategy; Graph Theory; Minimum Spanning Tree; Depth-First Search

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# 1 Introduction

## 1.1 Background

As modern architecture employs increasingly complex structures and layouts for aesthetic purposes, the pragmatic risk of its intricate structure lies in an elevated difficulty of evacuation. In recent years, the rising number of casualties due to failure in evacuating highlights the urgent need for more effective evacuation strategies, most notably during emergencies such as fires, gas leaks, etc. These failures are extremely dangerous and can lead to fatalities through crushing, as crowds can create physical pressures that reach 4,450 N/m, sufficient to bend steel barriers [1]. Under such extreme conditions, self-organized evacuation will be unsuccessful, therefore creating environments where untrained individuals are incapable of overcoming the chaotic, jammed crowd. This makes the intervention of trained personnel (responders) who can manage the flow and guarantee the successful evacuation of the crowd necessary and crucial.

The fact that orderly evacuation rarely occurs in a crisis demonstrates the critical role of responders. Past research into human behavior during fires has revealed that occupants rarely begin evacuation immediately upon alarm, but they seek information and gather their belongings before they actually attempt to leave the building [2]. This period of delay can be fatal. Moreover, without clear instruction and guidance, congestion and other accidents may cause people to overlook alternative exits or escape routes in panic. Therefore, not only are responders necessary for successful evacuation, but a strategy must be established in order for the responders to conduct an efficient evacuation sweep. However, the diversity of building designs and types of emergency make it difficult to develop a universal evacuation strategy. A one-size-fits-all approach fails in this situation.

During real emergency conditions, practical obstacles such as communication delays, inefficient resource allocation, and public panic often hinder evacuation success and can result in high deaths tolls. Panic can induce a behavior paradox of the "faster-is-slower" effect, where individuals' attempts to move faster than a critical threshold (approximately 1.5 m/s) lead to congestion at bottlenecks, ultimately slowing down evacuation [1]. Crowds' "mass behavior" exacerbates this situation since individuals tend to jam at one exit [3]. This typical human behavior during evacuation can lead to a failure in resource allocation (exits, routes, etc.), which lead to more deaths [4]. Moreover, responders sweeping buildings face the challenge of limited time and real-time pressure. These factors make the development of a systematic evacuation strategy crucial for minimizing casualties and improving evacuation outcomes.

Mathematical modeling offers a robust framework for this issue. By quantitatively representing building structure and responder behavior, models are able to simulate evacuation situations and therefore optimize evacuation routes and improve resource allocation. In this paper, we aim to construct a generalized solution for evacuation by identifying optimal strategies for responders to conduct sweeps and guarantee all-clear throughout the building in minimal time.

## 1.2 Problem Restatement

This study aims to develop an effective model for evacuation sweep operations. Therefore, it is necessary to consider the various factors that influence responder strategies and impact evacuation outcomes. The main objectives of this study are presented as follows:

- **Identify key factors in different emergency conditions:**

To maximize the model's effectiveness, it must account for critical variables that change

across different situations. Namely, the type of emergency and the building layout significantly impact the strategies responders should employ. Occupant behavior, such as evacuation speed and reaction to instructions, also affects evacuation efficiency. Categorizing these different emergency conditions provides a scientific baseline for assessing each scenario, leading to appropriate responding strategies.

- **Determine responder roles and capabilities:**

Responding effectively to different emergency conditions requires varying numbers of personnel. The model must define these roles and capabilities to ensure that the simulated strategies are able to reflect realistic limits. This includes modeling how personnel can divide and coordinate different tasks.

- **Map building layouts and constraints:**

The model must accurately represent distinctive building layouts and physical constraints, as these directly influence evacuation outcomes. A precise mapping of the physical space is essential for the evaluation of viable sweeping paths and optimal evacuation routes.

- **Model occupant behavior and movement:**

Occupants should be categorized into different groups based on their mobility and ability to evacuate efficiently. By considering these differences in speed and behavior, the model is able to predict potential congestion points and therefore strategically prevent bottlenecks at critical locations within the building.

- **Increased complexity of building layouts:**

When the evacuation involves multi-story buildings with varied floor plans and occupant distributions, these spatial characteristics can significantly influence evacuation outcomes. The model must be capable of determining the optimal sweeping strategy to maximize efficiency in these complex environments. Estimating the shortest time required for a thorough sweep is necessary to evaluate the overall success of the evacuation process and the effectiveness of the model.

- **Expanding the model to include more realistic constraints:**

Incorporating additional real-world constraints, such as communication delay and public panic, will enhance the universality and effectiveness of the model. The evaluation of selected additional factors should enable the model to provide more accurate estimate of sweeping time for more complex situations, demonstrating the model's potential for broader application.

## 2 Assumptions and Justifications

To simplify the problem of analyzing and quantitatively assessing emergency evacuation sweeps, we make our assumptions as follows.

- **Assumption 1: Systematic Visual Scanning Time**

We assume that it takes a definite amount of time to confirm occupants in the room, which includes the time of a basic, visual scan and a more thorough examination that guarantees an empty room.

- **Justification 1:**

While the time to do an immediate visual scan is short, the time required to guarantee no occupant presence is significant because responders need to check throughout the room. Therefore, a certain amount of time, including a basic scan and a final confirmation, is required to ensure the room is empty.

- **Assumption 2: Constant Responder Speeds and Search Efficiencies**

We assume that the responders normally sweep a building with a certain speed, if confounding factors are not present. These confounding factors include smoke during fire, congestion due to chaos, etc. In general conditions, responders have equal speeds and search efficiencies, while their speeds and efficiencies may vary among evacuation conditions, depending on the type of emergency. We do not assume that under all emergency conditions and at all stages of evacuation, responder speeds and search efficiencies remain constant.

- **Justification 2:**

Responders are generally from the same organization. For instance, if the responders are firefighters, they probably come from the same station. Consequently, they receive similar training and should have similar speeds during the event. During different types of emergencies, however, there might be adverse conditions that hinder their speeds. This also causes them to have similar search efficiencies (i.e., they can search an approximately equal area during an equal period of time), which, likewise, differs due to the types of emergencies.

- **Assumption 3: Unusable Doors Due to Emergency**

During a particular evacuation event, we assume that the doors are in a good state unless the emergency renders them malfunctioning, and it takes more time for responders to enter the room if the door is not working properly. The proportion of doors in a bad state may vary based on the type and magnitude of the emergency.

- **Justification 3:**

Doors break due to emergencies when, for instance, earthquakes render the doors out of shape, thus unusable. Natural depreciation is not taken into account as the equipment, including doors, is usually regularly renewed to maintain usage.

- **Assumption 4: Knowledge of Layout and Structure**

We assume that the responders have access to the building's layout and structure, including positions and dimensions of rooms, hallways, and corridors on all floors, and the layout and structure have not significantly changed afterwards, so that the knowledge is accurate.

- **Justification 4:**

Usually, responders contact the owner(s) of the building to look for the layout and structure, and the data is almost always accessible. Having data that accurately reflects the layout of the rooms, hallways, and exits' positions and dimensions is the prerequisite for preparing an optimal strategy.

- **Assumption 5: Negligibility of Fine-Scale Structures in Layout Mapping**

We assume that fine-scale structural and non-structural elements can be neglected when mapping a building's layout. This includes, but is not limited to:

- The finite width of interior walls and partitions.
- The cross sectional area occupied by pillars and support columns.
- Minor furnishings such as desks, chairs, bookcases, and others depending on the specific situation.

• **Justification 5:**

The cumulative area occupied by these structures is negligible compared to the gross area of a room or corridor. Since the model's primary purpose is to predict sweeping strategy under dangerous conditions, the omission of these minor structures does not significantly impact the accuracy of these predictions. Moreover, in a high-stress emergency response scenario, responders conduct a rapid visual assessment to understand the overall layout and locate occupants. Therefore, the model should prioritize the identification of room connectivity, exit routes, and large obstructions instead of fine details. These details are secondary to the overall space for navigation.

### 3 Variables

We summarize the variables used in the following essay, together with their corresponding definitions or explanations.

Table 3.1: Variables Table

| Variables      | Denotation, Definition, and/or Description                                        |
|----------------|-----------------------------------------------------------------------------------|
| $v$            | Vertices or Nodes (Rooms, Exits, Parts of Hallways)                               |
| $e$            | Edges (Connections Between Vertices)                                              |
| $s$            | Speed of Responders (m/s)                                                         |
| $\alpha$       | Smoke Coefficient                                                                 |
| $V_i, S_i$     | The Minimum Spanning Tree's $i$ -th Subtree's Vertices and Edges, respectively    |
| $T_{door}(v)$  | Time to Open the Door to Room $v$ (s)                                             |
| $T_{clear}(v)$ | Time to Clear the Room $v$ (s)                                                    |
| $T_{trav}(e)$  | Time to Travel Through the Edge $e$ (s)                                           |
| $T_{room}(v)$  | Time to Sweep the Room $v$ , Including Entering, Clearing, and Leaving (s)        |
| $T_{total}$    | Total Time to Sweep the Entire Building (s)                                       |
| $T(S_i)$       | Total Time to Sweep the Subtree $S_i$ (s)                                         |
| $w, l$         | Width and Length of a Room or Hallway(m)                                          |
| $\rho$         | Parameter of Visibility, the proportion of immediately visible area to total area |
| $\eta$         | Search Efficiency (the area searched in detail during unit time) ( $m^2/s$ )      |

## 4 Modeling

### 4.1 Sweeping Strategy in a Single-floor Office Building

We first consider the case of a one-story building consisting of six offices, three on each side of a hallway connected to two exits on each side. In Figure 4.1, we present the floor plan as a graph where nodes represent rooms, exits, and portions of the hallway, and edges represent doors or connections between portions of the hallway.

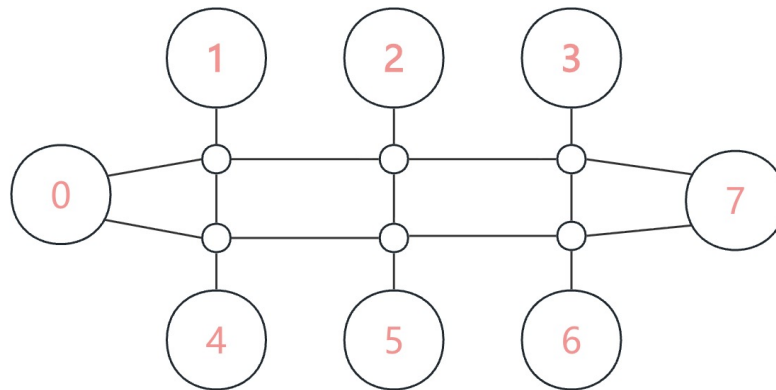


Figure 4.1: Floor Plan Graph

Nodes 0 and 7 represent exits through which the evacuating crowd can leave, and nodes 1 through 6 are rooms requiring search. Disregarding time-sensitive events, the time required to check the rooms and guarantee emptiness is roughly the same regardless of the order; what is left is only the sweeping order.

Since we need to minimize total time, two firefighters must each sweep three rooms to balance workload; in doing so, there are no significant differences in required time, and they will finish work about the same time. In addition, they would eventually evacuate from either exit, so it takes more time if they evacuate from the same exit where they entered, since they are repeating their previous paths without sweeping any additional rooms.

Therefore, they must follow a sequence of 0-1-2-3-7 (or reverse) and 0-4-5-6-7 (or reverse) to traverse the building. This sweeping strategy can be systematically implemented using a Depth-First Search (DFS) approach on the graph representation of the building. For example, firefighter A, following (say) path 0-1-2-3-7, operates as the following DFS approach:

- Start at node 0 (exit)
- Move to node 1, perform room clearance ( $T_{room}$ )
- Continue to node 2, perform room clearance
- Proceed to node 3, perform room clearance
- Finally exit through node 7

DFS is an algorithm widely employed for finding solutions to problems in graph theory and artificial intelligence. When selecting an edge to go through, the DFS will choose the edge coming out of the vertex that we have most recently visited [5]. The DFS algorithm ensures that once a room is visited and cleared, the firefighter moves to the next unvisited room in the deepest possible path before backtracking, thus providing an efficient sweeping order. This approach guarantees complete coverage while providing an optimal sweeping strategy that minimizes total time. The DFS aligns well with our optimization goals, as it systematically explores all vertices (rooms) in a portion of the building while maintaining a linear time complexity of  $O(V + E)$ , where  $V$  represents rooms and  $E$  represents connections between them.

Employing the sweeping strategy above, we proceed to estimate the time needed to clear the building. In the following calculation, we assume that the length ( $l$ ) and width ( $w$ ) of each office are 7 meters by 6 meters, respectively, and the width of the hallway is 3 meters. As they are offices, most of the places are immediately visible, so they have relatively high coefficients of visibility ( $\rho$ ) at around 0.8 – 0.9, and the search efficiency ( $\eta$ ) is roughly 0.5 – 1.5 m<sup>2</sup>/s. Therefore, the time required to clear a particular room is

$$T_{clear} = \frac{wl(1 - \rho)}{\eta}, \quad (4.1)$$

Equation 4.1 demonstrates the relationship between the estimated time to clear a room and parameters like the room's area, visibility, and the responders' search efficiency. It shows that the required time is directly proportional to the area and negatively associated with both visibility and search efficiency.

Plugging in the mean of the estimated values ( $\rho = 1, \eta = 1 \text{ m}^2/\text{s}$ ) yields

$$T_{clear} = \frac{7 \times 6 \times (1 - 0.85)}{1} = 6.3\text{s}$$

Assuming all doors are properly functioning, and it takes about half a second to open a normal door, then

$$T_{room} = T_{open} + T_{clear} = 0.5 + 6.3 = 6.8\text{s}$$

Therefore, the total time required is equal to the sum of time to complete sweeping the rooms and the sum of time to travel in between rooms; assume that the speed of the responders is constantly 2 m/s on average, then each firefighter has to go through two segments connecting two rooms (7 m) and two segments connecting an exit and a room, which is the hypotenuse of a right triangle that has half the corridor width and half the room length as legs ( $\sqrt{(\frac{1}{2} \times 7)^2 + (\frac{1}{2} \times 3)^2}$  m). As the firefighters are traversing the building in parallel, we only have to calculate the time it takes for one of them to finish:

$$T_{total} = 3T_{room} + T_{trav} = 3 \times 6.8 + \frac{2 \times 7 + 2 \times \sqrt{(\frac{1}{2} \times 7)^2 + (\frac{1}{2} \times 3)^2}}{2} \approx 31.2079\text{s}$$

This means that we need approximately 31.2 seconds to clear the entire building to guarantee that nobody is left behind. This model, while simple, uses DFS to strategically manage the sweeping process, providing a basic model upon which further model development is made possible.

## 4.2 Generalized Model For More Complex Layouts

While the DFS-based sweeping strategy is proven effective and efficient in our previous single-story building, real-world environments introduce new layers of complexity. Multi-floor structures, irregular room layouts, and varying exit placements require a more systematic approach that cannot be fully addressed by a simple traversal strategy. To better manage these realistic building configurations, we extend our model by incorporating Minimum Spanning

Tree (MST) partitioning, which enables balanced task allocation and reduces global sweeping time through multi-agent cooperation.

The Minimum Spanning Tree (MST) is a fundamental concept in graph theory that selects a subset of edges connecting all vertices in a weighted graph with the minimum total edge cost while avoiding cycles [6]. For sweeping and search operations, the MST represents the most economical connectivity structure of the floor plan, ensuring that responders traverse only essential connections while still maintaining full coverage. Among the classic algorithms used to construct the MST are Prim's algorithm, which grows the tree from a starting vertex by repeatedly adding the cheapest outgoing edge, and Kruskal's algorithm, which sorts all edges by weight and incrementally adds them if they do not form a cycle [7].

In our evacuation-planning model, we adopt Kruskal's algorithm rather than Prim's for two key reasons. First, building floor plans typically produce sparse graphs, where the number of edges is close to the number of vertices. In such cases, Kruskal's algorithm which sorts edges globally and uses an efficient union-find structure performs better than Prim's matrix-based variant and competes favorably with heap-optimized implementations [9]. Second, Kruskal's algorithm naturally handles initially disconnected or partially separable regions, which frequently occur in real architectural layouts due to closed rooms, separated zones, or restricted-access hallways [9]. This property allows responders to easily identify isolated components and understand which areas are reachable or potentially blocked. Consequently, Kruskal's algorithm offers both computational efficiency and practical utility for constructing MST in complex, multi-floor buildings.

The integration of MST ensures that each responder follows the most efficient route while covering the entire building. By partitioning the whole area into different sections and assigning them to respective responders who perform sweeping tasks simultaneously, each portion of the building is swept with the assistance of the MST theory. We thereby identify an optimal strategy that minimizes time.

#### 4.2.1 MST-Based Partitioning Strategy

The core of our generalized approach includes constructing a Minimum Spanning Tree from the building graph and dividing it into balanced subtrees assigned to individual responders while balancing workload, thereby minimizing total time. The specific methodology works as follows:

1. First, we simplify the problem and represent the building as a weighted graph  $G = (V, E)$  where vertices represent rooms, exits, and key hallway points, and edges represent connections with weights corresponding to travel times.
2. We then apply Kruskal's algorithm to construct a Minimum Spanning Tree that connects all critical nodes with minimal total edge weight, ensuring overall connection.
3. Afterwards, we finally partition the MST into  $k$  connected subtrees that use a load-balancing algorithm that minimizes the maximum sweeping time among all responders. That is, we are looking for

$$\min_{i=1,2,\dots,k} \max W(S_i) = \min_{i=1,2,\dots,k} \max \left( \sum_{v \in V_i} T_{clear}(v) + \sum_{e \in S_i} T_{trav}(e) \right), \quad (4.2)$$

where  $S_i, i = 1, 2, \dots, k$ , forms a partition of the minimal spanning tree, represents the total sweeping time for subtree  $S_i$ , and the minimum is taken over all possible  $k$ -subtree partitions of the MST. The weight,  $W(S_i)$  comprises both room clearance times and travel times within the subtree  $S_i$ .

4. Finally, assign each connected subtree to a responder while ensuring that:

- Each subtree is connected, and connects with at least one exit;
- The maximum sweeping time among all responders is minimized;
- All rooms are covered in at least one subtree.

### 4.3 Application to Real-World Structure: The White House

Having established a generalized model with MST partitioning, we now apply it to a real-world building: the White House. The White House, with its two floors and complicated floor layout, provides an ideal testing case to appraise the effectiveness of our model, as shown in Figure 4.2.

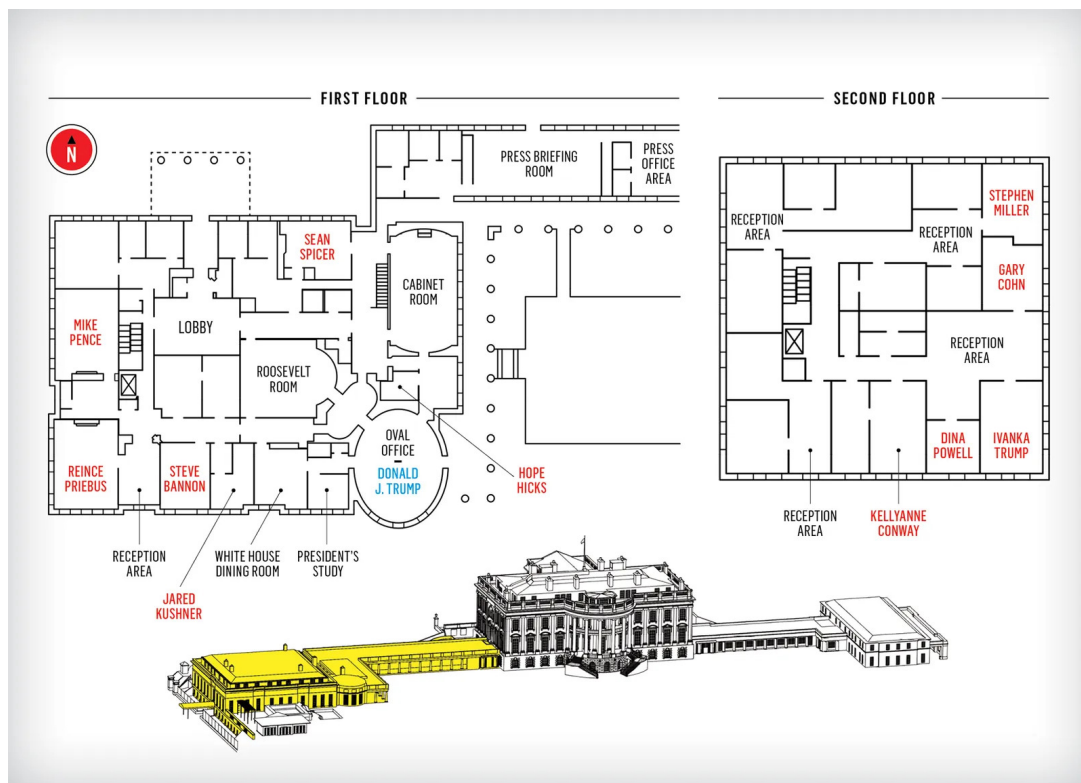


Figure 4.2: The White House [10]

For the White House, we construct a weighted graph consisting of 43 rooms across two main floors, where edge weights reflect travel times. The MST partitioning algorithm then optimally distributes these rooms among available responder teams. In Figure 4.3, the rooms and exits are labeled 1 through 43 in ocean-blue colored nodes, connecting hallways are labeled A through L in white colored nodes, the lift is labeled I and II (representing the floor it is on) in dark-blue colored nodes, and nodes are connected to each other via edges, with numbers representing weight (travel time).

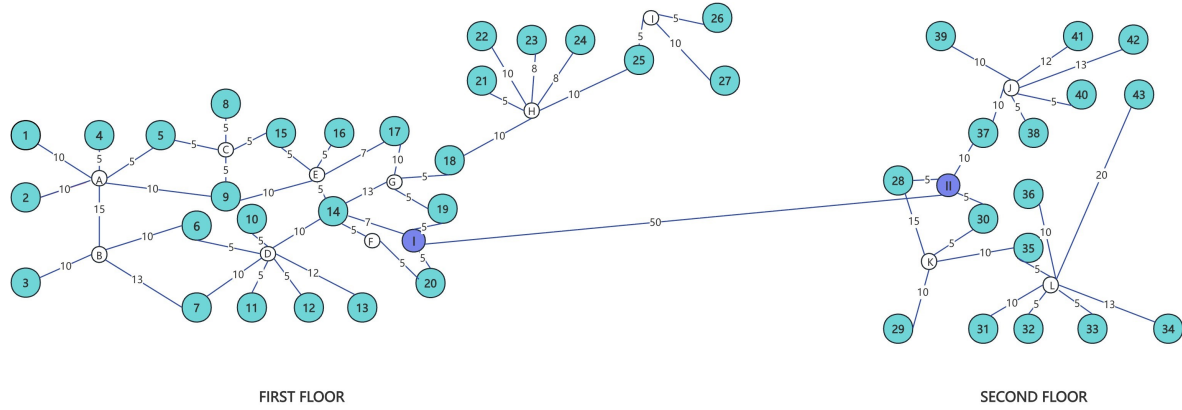


Figure 4.3: The White House (Graph)

In this subsection, we assume that it takes approximately the same time to search each room and all doors are working properly, and we use the value of clearance time we obtained in the previous section; that is,  $T_{clear} = 6.8$  s. We consider the evacuation procedure where there are two responders with a constant speed of 2 m/s, both of whom begin from and end at Node 8. Implementing our MST partitioning strategy, a simple employment of code yields that at least 520.8 s is required to sweep and clear the building. Given below is the rooms each responder will search and the order in which they search these rooms, according to our strategy.

- **Responder 1:**

Total Time: 520.80 s

Complete Path (23 nodes): F1 room08 - F1 room16 - F1 room14 - F1 room20 - F1 room17 - F1 room10 - F1 room11 - F1 room03 - F1 room04 - F1 room24 - F1 room23 - F1 room25 - F1 room26 - F1 room27 - F2 room30 - F2 room33 - F2 room32 - F2 room34 - F2 room43 - F2 room40 - F2 room41 - F2 room42 - F1 room08

- **Responder 2:**

Total Time: 449.80 s

Complete Path (23 nodes): F1 room08 - F1 room05 - F1 room09 - F1 room15 - F1 room01 - F1 room02 - F1 room12 - F1 room06 - F1 room07 - F1 room13 - F1 room19 - F1 room18 - F1 room21 - F1 room22 - F2 room28 - F2 room37 - F2 room38 - F2 room39 - F2 room35 - F2 room36 - F2 room31 - F2 room29 - F1 room08

This real-world application demonstrates how our model is not only theoretically sound but also capable of dealing with complex architectural environments. By employing robust algorithms and incorporating necessary, reasonable assumptions, the model achieves both computational efficiency and practical usefulness.

#### 4.4 Application to Real-World Structure: School Building

In addition to the White House scenario, we also apply our model to a five-story school with an L-shaped floor layout. This complex, multi-floor structure with rooms of different sizes

requires more responders, thus providing an ideal opportunity for the evaluation of the model's performance in an even more complex environment. The five floors are composed of different rooms, respectively:

- First Floor (Figure 4.4): Auditorium hall, Lobby, Washroom
- Second-Fourth Floor (Figure 4.5): Teacher's office, Classrooms, Washroom
- Fifth Floor (Figure 4.6): Library, Classrooms, Washroom

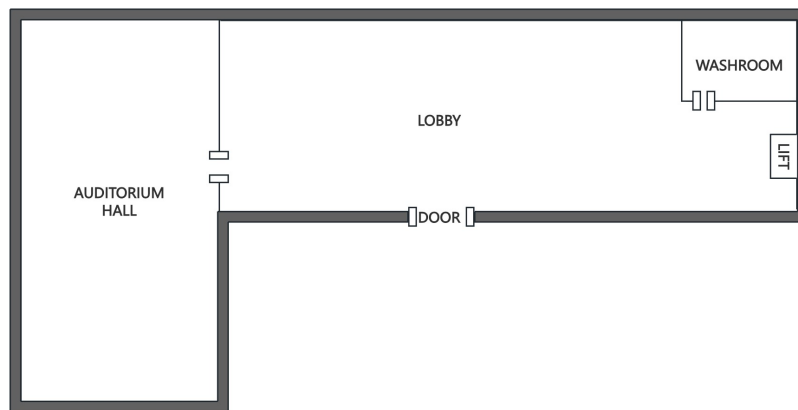


Figure 4.4: First Floor

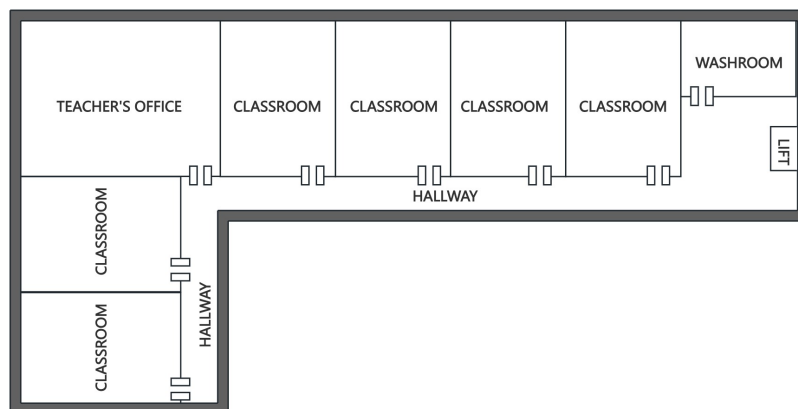


Figure 4.5: Second to Fourth Floor

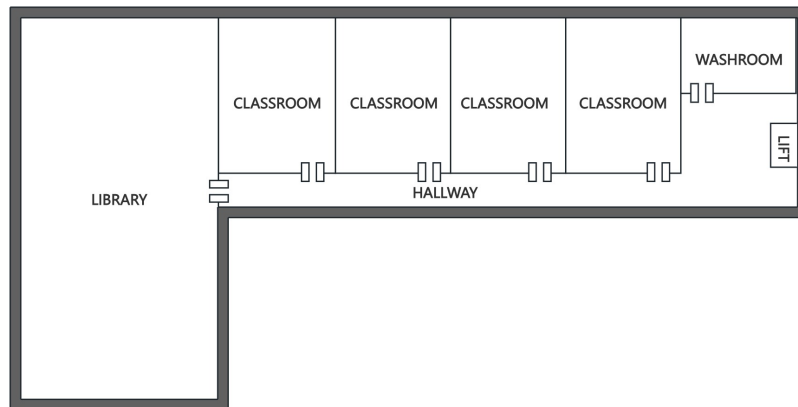


Figure 4.6: Fifth Floor

For the school, we construct three weighted graphs representing the five floors, where edge weights reflect times required. Similar to the White House scenario, the MST partitioning algorithm distributes the rooms and floors among responders. In the following figures representing different floors, different types of rooms are labeled with different colors, and alphabets are used to label all the rooms and the lift. Junctions are labeled with white-colored nodes and numbers, and they are interconnected via edges.

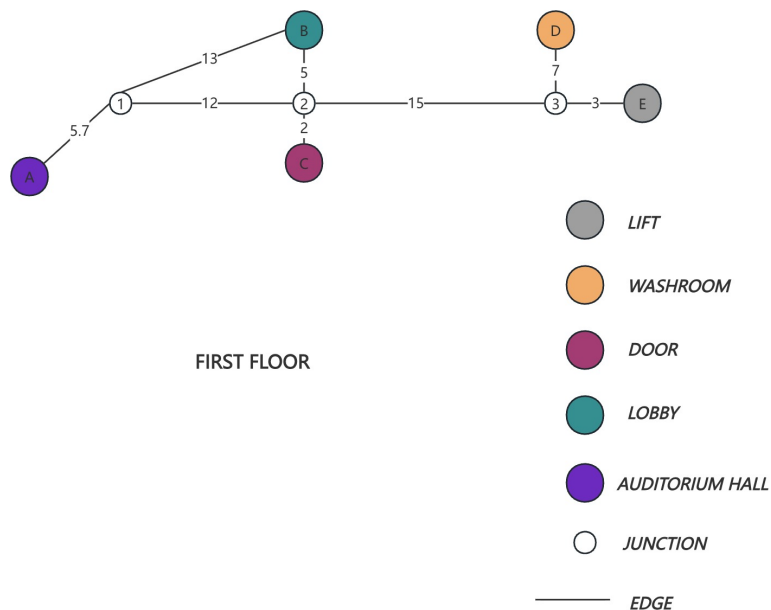


Figure 4.7: First Floor (graph)

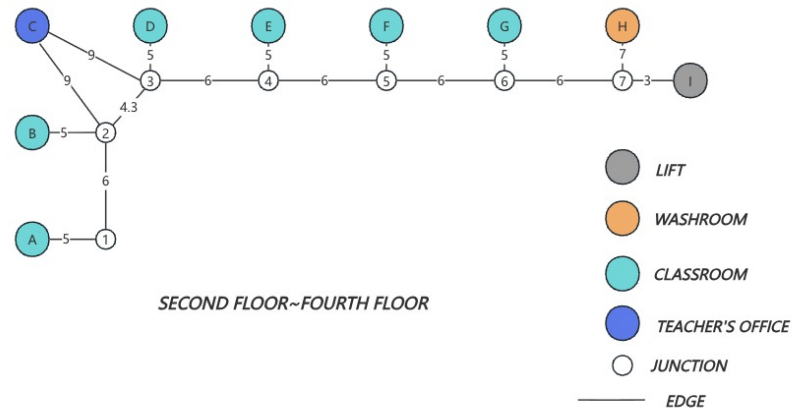


Figure 4.8: Second to Fourth Floor (graph)

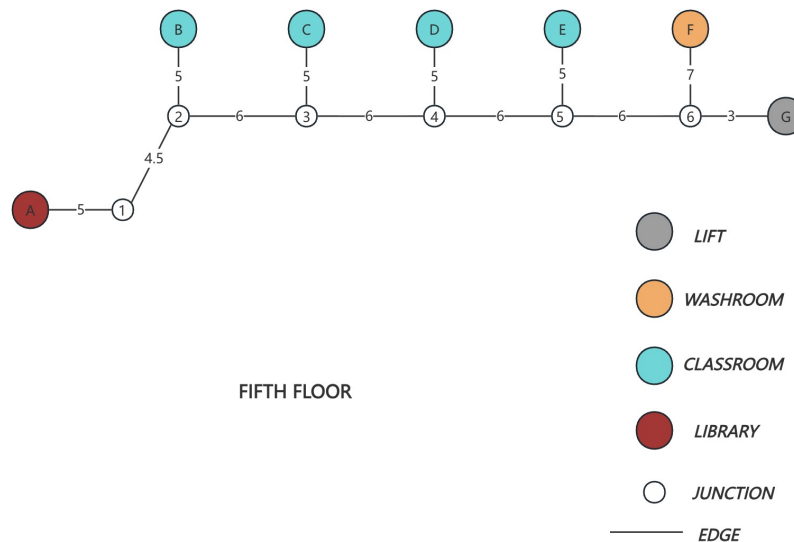


Figure 4.9: Fifth Floor (graph)

In a school setting, we consider an evacuation procedure of assigning 5 responders each with a constant speed of 2 m/s. All of the responders begin from and end at node C, which is the door, on the first floor to ensure safe evacuation of all involved. Knowing all the conditions and resources available, our model gives the result that at least 383.1 s is needed to completely sweep and clear the building, followed by a safe evacuation. Given below is the rooms each responder will search and the order in which they search these rooms.

- **Responder 1:**

Total Time: 275.80 s

Complete Path (10 nodes): F1 room1C - F2 room2G - F3 room3F - F3 room3A - F5 room5B - F4 room4D - F4 room4G - F2 room2I - F1 room1D - F1 room1C

- **Responder 2:**

Total Time: 255.00 s

Complete Path (10 nodes): F1 room1C - F1 room1B - F5 room5C - F5 room5A - F4 room4I - F3 room3H - F3 room3G - F2 room2E - F2 room2F - F1 room1C

- **Responder 3:**

Total Time: 383.10 s

Complete Path (10 nodes): F1 room1C - F1 room1A - F3 room3C - F4 room4A - F4 room4B - F5 room5G - F5 room5E - F2 room2C - F2 room2A - F1 room1C

- **Responder 4:**

Total Time: 227.60 s

Complete Path (9 nodes): F1 room1C - F4 room4F - F4 room4H - F5 room5F - F3 room3D - F3 room3B - F2 room2B - F1 room1E - F1 room1C

- **Responder 5:**

Total Time: 242.80 s

Complete Path (9 nodes): F1 room1C - F5 room5D - F4 room4C - F4 room4E - F3 room3E - F3 room3I - F2 room2D - F2 room2H - F1 room1C

This illustrates the model's capability of handling evacuation sweep of multi-story buildings, successfully allocating responder resources to minimize sweeping time and advise evacuation strategy. The model demonstrates its usefulness in schools, a setting where evacuation is often led by inexperienced teachers followed by juniors if responders are not present. Even if responders are present, the lack of a correct strategy can result in severe consequences. Compared to the White House, where experienced personnel are constantly on call, the school is a much more fragile place and needs an evacuation model even more. Therefore, the outstanding performance of the model in a school setting broadens the scope of its future applications.

## 5 Incorporating More Realistic Factors

The application on both the White House and the school building demonstrates the model's capacity to systematically and strategically assign specific instructions for responders to take action, which provides basic guideline for rapid evacuation. However, our previous examples are subject to changes caused by other factors in realistic emergency conditions. For instance, we cannot assume constant clearance time among different rooms and stable responder speeds during fire, as the smoke can be suffocating and urge rapid evacuation. Therefore, the model's strategy can be challenged by real-world problems that naturally occur in emergencies. Addressing these challenges, such as fire spread, presence of smoke, or occupants' movements, enhances the accuracy of the model and implementability of the strategies it provides.

In this section, we improve the previous model by taking into account more complex factors that otherwise alter the optimal result, thereby producing a more reasonable and actionable sweeping strategy. Specifically, we consider three frequent challenge in emergencies: fire spread and areas with higher priority. In doing so, the model can be applied to a broader range of incidents in real-life situations, instead of remaining a theoretical approach to predicting evacuation methods. Thus, it is a transformation from data-based prediction to pragmatic, actionable guidance.

## 5.1 Factor 1: Fire Spread

During a fire incident, fire inevitably spreads from its origin until it covers everything in the building. The spread of fire is a key factor in such an incident, as it not just renders areas inaccessible (for example, elevators), but also creates smoke that suffocates people nearby and lowers visibility of the surrounding environment[11]. Eventually, both the responders and occupants slow down.

Consider a fire emergency, where the fire originates from one of the rooms and spreads throughout the entire building. As the fire generates smoke when spreading, the responders sweep much more slowly since they need to search rooms with more caution in case the heavy smoke hinders their vision and eventually leads to undiscovered occupants. That is,  $s_{real} = \frac{s}{\alpha}$  for a constant  $\alpha > 1$  (called the smoke coefficient) where smoke is present. Furthermore, the actual time required to finish checking a room or traveling between rooms,  $T_{real}$ , might increase due to a decrease in speed and visibility. The actual time needed can be calculated using the formula below.

$$T_{real,room} = \begin{cases} T_{room}, & \text{if smoke is not present,} \\ \alpha T_{room}, & \text{if smoke is present,} \end{cases} \quad (5.1)$$

and

$$T_{real,trav} = \begin{cases} T_{trav}, & \text{if smoke is not present,} \\ \alpha T_{trav}, & \text{if smoke is present.} \end{cases} \quad (5.2)$$

We use a dynamic process that accounts for the spread of the fire by having a constant spread rate of the smoke, and whichever areas with the smoke are impacted are accounted for by the multiplication of  $\alpha$  as in the formula above.

The parameter,  $\alpha$ , represents the effect of the smoke on the sweeping process, ensuring a more reliable value of clearing and traveling time. By implementing the smoke parameter, we extend the use of our model by covering a crucial factor in fire emergencies.

## 5.2 Factor 2: Areas of Priority

In reality, not all rooms deserve the same amount of attention. Some areas have more priority than others; for instance, in a children's playground, a simple check might not always work. Since children are immature and lack the ability to make sound judgments, they might not be able to evacuate and might even enter rooms that responders have previously marked as clear. Hence, in an emergency that occurs at such places, some crucial areas demand more attention than others, so double- or even triple-checking on such areas is necessary.

Another example is a chemical factory, where large amounts of explosive chemicals are present. Since any mistake regarding such dangerous matter is deadly and may cause casualties to increase dramatically, responders need to pay extra attention to guarantee that nobody is in or nearby the area that deserves special treatment, in this case, where chemicals are present.

We address this problem by labeling each room either high-risk or low-risk. For low-risk areas, no special treatment is needed; however, for high-risk areas, we alter the algorithm by a little bit, so that the high-risk areas are visited and cleared at least twice, with a period of time in

between the two visits by responders to double-check clearance. We plan that prioritized zones are swept again at least 60 s to make sure a responder would not enter the room, and then leave, and then enter the room again, which is absurd and does not add to the point of double-checking as he might make the same mistake yet another time.

### 5.3 The White House Case Using Improved Strategy

Using the improved strategy described above, we reexamine the White House case, which provides a valuable testing ground. Recall that the White House is a two-story building with a total of 43 rooms, as shown below. In the following, we carry out an evacuation plan that follows our previous assumptions (i.e., two responders each with  $v = 2 \text{ m/s}$ ,  $\eta = 1 \text{ m}^2/\text{s}$ , and each room is a rectangle with  $l = 7 \text{ m}$ ,  $w = 6 \text{ m}$ ), as shown in figures 4.2 and 4.3.

Indicating rooms 9 and 20 as rooms with high priority, and simulating a fire that starts at room 14, we incorporated the factors discussed above into our strategy. The spread of smoke is assumed to be  $1 \text{ m/s}$  horizontally and  $6 \text{ m/s}$  vertically, and areas with smoke exhibit a smoke coefficient  $\alpha = 1.5$ . With this model, we found that two responders would need 616.60 s to search the entire White House. Given below is the order in which these two responders should search the rooms.

#### Responder 1:

Total Time: 616.60 s

Complete Path (25 nodes): F1 room08 - F1 room09 - F1 room05 - F1 room02 - F1 room01 - F1 room16 - F1 room09 - F1 room14 - F1 room20 - F1 room19 - F1 room21 - F1 room27 - F1 room20 - F1 room06 - F1 room07 - F1 room03 - F2 room29 - F2 room32 - F2 room33 - F2 room31 - F2 room36 - F2 room43 - F2 room38 - F2 room39 - F1 room08

#### Responder 2:

Total Time: 542.20 s

Complete Path (23 nodes): F1 room08 - F1 room15 - F1 room17 - F1 room10 - F1 room11 - F1 room12 - F1 room13 - F1 room04 - F1 room18 - F1 room23 - F1 room24 - F1 room22 - F1 room25 - F1 room26 - F2 room30 - F2 room28 - F2 room37 - F2 room40 - F2 room41 - F2 room42 - F2 room35 - F2 room34 - F1 room08

With the help of our model, despite accounting for two extra prioritized rooms and the spread of smoke, the total evacuation time only changed from 520.8 seconds to 616.6 seconds (a roughly 18% increase). Our model proves effective in incorporating key factors that naturally appear in the complexity of real-world situations, maintaining an efficient sweeping strategy.

### 5.4 Impact of Technology

As our previous analysis suggests, the model depends little on technology such as communication tools. While traditional search process requires high extent of cooperation among team members, our model assigns responders to different zones in which they sweep independently, reducing the dependence on communication tools. This is especially important as communication might fail during an emergency, which in turn hinders the evacuation. Our model, providing a thorough strategy, reduces the typical high dependence on technology by analyzing and preparing beforehand.

Nonetheless, advanced technology, such as sensors and cameras, assists the evacuation pro-

cess and reduces the overall required time. By providing useful information regarding occupants' number and positions, technological equipment identifies the crucial rooms – rooms where occupants are more likely to be present – and these areas, marked as high-risk, for instance, can be cleared with extra caution; on the other hand, areas identified as low-risk could be removed from the graph, simplifying the MST as a result.

Overall, while our model generally depends lowly on technology as it is a pre-planned strategy instead of evolving for the duration of the emergency, the presence of technology identifies occupants, which significantly reduces the total time by helping us remove unnecessary nodes on the graph representing the building.

## 6 Evacuation Sweep Strategy Overview

In this study, we employ a combination of DFS and MST algorithms to construct a generalized, robust, and effective evacuation sweep model to minimize sweeping time and improve evacuation outcome, as shown in Fig. 6.1.

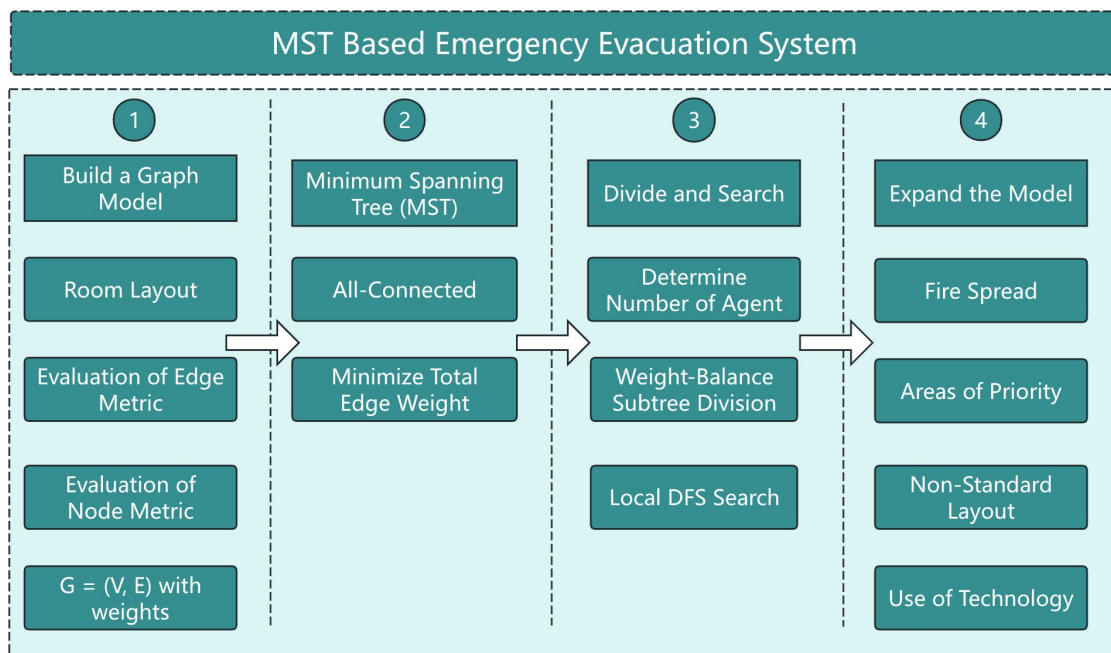


Figure 6.1: Strategy Overview

This overview figure illustrates our Minimum Spanning Tree (MST)-based Emergency Evacuation System, organized into four sequential task. In Task 1, we construct the underlying graph model. We start planning the model based on assumptions that state definite amount of search time, homogeneous responder groups, and knowledge of building layout based on the negligibility of fine-scaled structures. Stemming from these assumptions, the model considers a single-story building consisting of six offices, three on each side and connected by a long, central corridor bridging the exits. To represent the layout quantitatively and ideal for mathematical processing, we assign numbers to each room, label each room with vertices and the corridor with edges. The use of the DFS algorithm enhances path simplicity, as it ensures that no backtracking is needed before all rooms are searched. By systematically exploring all rooms and maintaining a linear time complexity of  $O(V + E)$ , the model calculates a cumulative time based on the

addition of sweeping time and traveling time, therefore obtaining an estimated value of 31.2 s to clear the single-story building. A use of DFS is proven sufficient and effective in proposing evacuation strategy for a simple single-story building, given all established assumptions.

Subsequently, due to the multi-story structure, varying room layouts, and different exit locations, an improvement of the model is strongly required to broaden its range of application and improve its real-world significance and implementability. Aiming to address this elevated spatial complexity and transcend the limitations of a simple traversal strategy, we bring MST algorithm into discussion. Task 2 generates a Minimum Spanning Tree (MST) from the weighted graph. The MST ensures global connectivity while minimizing total edge cost, providing an efficient backbone structure upon which evacuation routes can be planned. Task 3 focus on applying divide-and-search strategies. We determine the number of available agents, partition the MST into weight-balanced subtrees, and conduct local Depth-First Search (DFS) within each subtree. This enables parallel, workload-balanced navigation and reduces overall evacuation time. Specifically, we employ Kruskal's algorithm to deal with partially separated regions and isolated elements that are common in real-world building layouts. We name this "MST-based partitioning strategy," stressing on the model's ability to provide effective solutions for evacuation in multi-floor settings. Essentially, we generate graphs to represent floor layout and then assign weight to each edge. Kruskal's algorithm produces an MST to connect critical nodes, given minimum total edge weight. A minimum total edge weight will produce the minimum sweeping time, as its gives the least distance the responders must travel in order to sweep all spaces. Hence, task 4 expands the model by incorporating realistic conditions, such as dynamic fire spread, priority areas, non-standard building layouts, and sensor-based technological inputs. These extensions enhance the adaptability of the system to real-world emergencies. We systematize a real-world, complex evacuation problem into the calculation of minimum weight, thus resolving the original problem using mathematical approaches.

## 7 Sensitivity Analysis

In our previous sections, while some parameters are accurate (such as  $w, l$ ), many others are only estimations and would likely differ in reality, due partly to difficulties in obtaining exact values: parameters like  $\rho$  and  $\eta$  often need an experiment to accurately determine. It is important to determine if our model is robust when parameters are slightly altered, so we carry out a sensitivity analysis in this section to analyze the effect of a change (up to 10%) in the estimated parameters, namely  $\rho$  and  $\eta$ , on the overall sweeping strategy and total time.

### 7.1 Sensitivity With Respect to the Visibility Parameter, $\rho$

The visibility parameter  $\rho$  exhibits varying sensitivity regarding clearance time in our evacuation model, changes being occasionally proportional and occasionally significant to variations in  $\rho$ . This relationship stems from the direct mathematical dependence of room clearance time on visibility:

$$T_{clear} = \frac{wl(1 - \rho)}{\eta},$$

which shows that the clearance time within a single room is directly proportional to  $1 - \rho$ . We consider two separate cases: one where  $\rho$  is large and another where  $\rho$  is small.

Areas with Large  $\rho$ 's appear where almost everything is within sight, for instance, an unoccupied room or an empty playground. These areas have  $\rho$ 's close to or even equal to 1. For calculation's sake, say that the value of  $\rho$  changes from 0.8 by up to 10%. Then,  $1 - \rho$  changes

from 0.2 to 0.12 or 0.28. Since  $T_{clear}$  is proportional to  $1 - \rho$ , the clearance time in a room changes disproportionately highly to the variation of  $\rho$ , with a change of  $\pm 40\%$ . The total clearance time, then, could be highly sensitive to little disruptions of the value of  $\rho$  in case it is large.

On the other hand, areas with little  $\rho$ 's appear when many obstacles are present, for example, a utility room where tools are all over the place. These areas' values of  $\rho$  are close to 0. Again, for calculation, say that  $\rho = 0.1$  and changes by up to 10%. Then  $1 - \rho$  changes from 0.9 to 0.89 or 0.91, which is only about  $\pm 1.1\%$ . This is another disproportionate, yet disproportionately low, variation with respect to  $\rho$ 's. Then, the total clearance time would be highly insensitive to slight alterations of  $\rho$ .

For typical buildings,  $\rho$ 's are often considerably high (which is only natural because normally many places can be seen immediately in a room). In our previous sections, for instance, we assumed a constant  $\rho = 0.85$ ; our sensitivity test shows that the model is highly sensitive in this case. However, since the value of  $\rho$  only affects the total clearance time ( $T_{clear}$ ) but not the traveling time ( $T_{trav}$ ). Consequently, despite a high percentage of change in room clearance time, this change is reduced by the constant traveling time between rooms (which, more often than not, occupies the majority of total time), ultimately producing a mild-to-low overall sensitivity of total evacuation time regarding the visibility parameter.

## 7.2 Sensitivity With Respect to Search Efficiency, $\eta$

The search efficiency parameter  $\eta$  also plays a critical role in calculating the time needed to clear a room, as illustrated by the equation:

$$T_{clear} = \frac{wl(1 - \rho)}{\eta} \quad (7.1)$$

This shows that the clearance time within a single room is inversely proportional to  $\eta$ . We analyze the sensitivity by considering two cases: one where  $\eta$  is large and another where  $\eta$  is small.

Large values of  $\eta$  correspond to highly efficient search operations, often due to well-trained responders. If  $\eta$  is initially large, say  $\eta = 2.0 \text{ m}^2/\text{s}$ , and it changes by up to 10%, then  $\eta$  varies from 2.0 to 1.8 or 2.2. Since  $T_{clear}$  is inversely proportional to  $\eta$ , a 10% decrease in  $\eta$  (to 1.8) increases  $T_{clear}$  by approximately 11.1%, whereas a 10% increase (to 2.2) decreases  $T_{clear}$  by about 9.1%. Thus, the clearance time is moderately sensitive to changes in  $\eta$  when  $\eta$  is large, as the clearance time is roughly proportionate as  $\eta$  varies by a small amount. (Worth noting, when  $\eta$  varies significantly, the roughly proportionate association fails to persist.)

On the contrary, small values of  $\eta$  occur in challenging environments, such as those with poor visibility, complex layouts, or responder fatigue. If  $\eta$  is initially small, say  $\eta = 0.4 \text{ m}^2/\text{s}$ , and it changes by 10%, then  $\eta$  varies from 0.4 to 0.36 or 0.44. A 10% decrease in  $\eta$  (to 0.36) increases  $T_{clear}$  by about 11.1%, while a 10% increase (to 0.44) decreases  $T_{clear}$  by approximately 9.1%. Although the percentage change in  $T_{clear}$  is similar to the large- $\eta$  case, the absolute change in time is more significant when  $\eta$  is small because  $T_{clear}$  is already high. For instance, if  $T_{clear}$  is 100 seconds at  $\eta = 0.4$ , a 10% decrease in  $\eta$  adds over 11 seconds per room, which can accumulate rapidly in multi-room buildings.

Thus,  $\eta$  exhibits moderate sensitivity regardless of the magnitude. While it is significant in low-efficiency conditions, it is less influential on total time than  $\rho$  in high-visibility areas;

similar to  $\rho$ , the model's strategy is unaffected by changes in  $\eta$ , as it influences time but not path planning. Accurate estimation of  $\eta$  is crucial for reliable time predictions, but the model is robust against  $\eta$  in terms of sweeping strategy planning.

## 8 Strengths and Weaknesses

By incorporating DFS and MST, the evacuation sweep model maps building layout and proposes optimal evacuation strategies that ensure the efficiency of the process and safety of both occupants and responders. Starting from a simple one-story building, the model uses DFS to propose a strategy that minimizes total evacuation time. We then incorporate MST to produce a more generalized model suitable for more complex layouts, drawing factors like multi-storied and variable floor layouts under consideration. This approach allows our model to align closely with the purpose of evacuation: minimize time cost and maximize survival. We finally conduct a thorough evaluation of the model to conclude its strengths and weaknesses.

### 8.1 Strengths

- **Broad Range of Application:**

The model addresses critical problems in current evacuation process, where inexperienced occupants easily miss direction during emergency conditions even with the guidance of trained responders. The model's ability to offer reasonable strategies addresses key issues in evacuations: low survival outcome and evacuation efficiency.

- **Comprehensiveness of Evaluation:**

In preparation for the model, we consider a wide range of factors that may interfere with evacuation outcomes, and these factors are of increasing complexity and with significant relevance. In doing so, the model enables a holistic, comprehensive assessment of emergency conditions and therefore generates a strategy that is generally the optimal solution.

- **Incorporation of Sensitivity Analysis:**

We do a sensitivity analysis by examining how slight changes in certain factors such as room search time will affect the accuracy of the strategy our model produces. The analysis of these factors adds to the robustness of implementability of our model, as it becomes more resistant to the fluctuation of certain factors due to real-life conditions.

- **Use of Robust Algorithms:**

The algorithms we employ in our model, mainly DFS and MST, are renowned models that have demonstrated their robustness and effectiveness in many previous works.

### 8.2 Weaknesses

- **Restraint of Computational Resources:**

Due to the large set of data we must process for the simulation of evacuation paths and minimum sweeping time, the computational resources currently available are insufficient in solving such NP-hard problems. We did our best to address multiple emergency conditions with increasing complexity and real-life significance.

- **Coincidental Factors in Reality:**

The model is fundamentally based on proposed building structures and generates prediction of emergency condition. However, the strategies it proposes can be affected by

various unpredictable factors in real-life emergency conditions, such as the usability of furniture and exits.

Despite the limitations above, our model nonetheless showcases remarkable value for planning the evacuation process: the integration of DFS and MST builds a sound foundation that makes it widely applicable in various emergency situations. While we acknowledge the restraints of computational resources and unpredictable changes in reality, the model's strength, especially its broad applicability, comprehensive account of multiple factors, and robustness against parameter variations, makes it valuable for emergency planning. In our future work, we would put emphasis on improving computation efficiency and putting even more dynamic factors into consideration, thereby enhancing the model's reliability in more diverse scenarios.

## 9 Conclusion

In this paper, we have developed and validated a robust computational model for optimizing evacuation sweep procedures in complex building environments. Our model integrated Depth-First Search (DFS) for efficient room-by-room clearance and Minimum Spanning Tree (MST) algorithms for path optimization and workload distribution among multiple responders. This strategic approach provides a pre-planned strategy that minimizes total evacuation time, thereby enhancing survival outcomes.

The core strength of our model is its adaptability and comprehensiveness. We start our study with a single-story layout for the establishment of a foundational framework, and then we adapt the model to address evacuation conditions in multi-story structures. This incorporation of real-world factors, including fire spread and zone of priority, demonstrates the model's ability to deal with hazardous conditions in real life. Several simulations reveal that the model's strategy lowers evacuation time effectively, confirming its practical usage.

Furthermore, we incorporate a sensitivity analysis that reveals the model's robustness. The total evacuation time exhibits a degree of sensitivity to the visibility parameter ( $\rho$ ) and search efficiency ( $\eta$ ), but the sweeping strategy is not significantly affected. Therefore, even if there are practical uncertainties, the pre-planned paths are still valid. The model's design that assigns different zones to individual responders also adds to its reliability in extreme situations where technology may fail.

Despite its strength, we acknowledge certain limitations, primarily the computational constraints in solving NP-hard problems for extremely large datasets. Moreover, the potential influence of unexpected real-world variables is always subject to change. In summary, this model presents a significant step forward in transforming evacuation planning from an improvised process into an optimized, mathematically-grounded strategy. It will contribute to safer and more efficient emergency responses.

## 9. A Letter to the Local Emergency Planning Committee

Dear Committee Members,

We would like to appreciate your continued, dedicated effort in safeguarding our community's well-being through robust emergency planning. Aiming to make practical contribution to our community, we are pleased to present our research on the optimization of evacuation sweep procedures. In an era where emergencies can escalate with high speed, the critical time during an evacuation often determines life and death. Our research team has dedicated its effort to a crucial phase of emergency response: the sweep of a building, where responders must ensure that every single occupant is located and has been successfully evacuated. This high-stakes, time-sensitive process can lead to chaotic situations, even with the most experienced personnel.

Current evacuation sweeps often relies on the improvised judgment of responders who are operating under huge stress and in hazardous conditions. Therefore, such approach can lead to unchecked areas in a building and the waste of efforts, eventually extending the time needed for a full evacuation. This can leave both occupants and responders at greater risk. Studies by the National Institute of Standards and Technology (NIST) on building evacuations demonstrate that pre-planned protocols significantly work better than situational decision-making. We believe there is a significant opportunity to enhance the efficiency and effectiveness of emergency responses through a systematic approach, thereby saving precious sweeping time and lives.

To address this existing challenge, our research conducted an analysis to improve emergency evacuation sweep procedures for community buildings. By applying a computational model to multiple architectural structures, including a single-story office and a five-story school, we have identified and verified that such systematic approach can significantly enhance survival outcomes by assigning responders and minimizing evacuation time, compared to improvised, often underprepared responses. For instance, in a simulated five-story school scenario, our strategy enabled a team of five responders to achieve a complete sweep and evacuation in 383.10 seconds, significantly shorter compared to improvised evacuation response.

The core of our proposal is a strategic framework built on three key principles, designed to integrate with existing emergency protocols:

### 1. Pre-planning and Path Assignment:

For high risk facility, we recommend creating an accurate map focusing on major structures and layout. If this map is provided, our model is able to process this map to generate efficient search routes for responders. Each responder should be assigned a specific path that guides the route of the sweep, thus increasing sweeping efficiency by minimizing sweeping time. It also ensures all-clear for the whole building, while preventing waste of effort as every room will be only covered once.

### 2. Identification of "High-Priority" Zones:

Identifying "high-priority" zones is essential for guaranteeing positive survival outcome. Searching these areas where occupants are most vulnerable or under greatest threat, such as factories and

schools, is a priority. The system addresses this issue by assigning high-risk or low-risk label to an area, thus scheduling a mandatory second sweep for high-risk zones. Therefore, this ensures that no occupant is left in the building and improves evacuation outcome. Being aware of different levels of risk among areas in a building is essential for a successful evacuation sweep operation.

### 3. Maintain Safety Through Adaptability:

The pre-planned paths represent the optimal result given by the model for the conditions offered to it. Therefore, it requires dynamic adjustments to ensure the safety of the responders, in case that there are unexpected accidents. If a team encounters an unexpected hazard like structural damage, they should immediately report it. Instructing responders to report such obstacles immediately allows for real-time rerouting and reassignment of tasks based on realistic conditions, thereby maintaining the sweep's integrity without compromising personnel safety. Our model considers factors like reduced visibility from smoke, while other factors that potentially interferes with evacuation sweeps need extra attention.

Based on this analysis, we strongly recommend that the Committee consider adopting this framework. Certain actionable recommendations would help for diverse settings, according to building types and occupant needs:

- **Schools:** The full framework mentioned previously should be implemented, as classrooms with younger students are among the most vulnerable. It also includes lots of spaces necessary for a double-check. Therefore, trained staff should first gather students in orderly groups as much as possible, and then the responders are able to sweep the building without distraction.
- **Offices:** Large, open areas need to be cleared quickly. But in this case, a single responder should be able to summon a long line of educated adults to leave, if they are close to the exits. Other responders can follow the strategy to complete the sweeping process. However, private offices and conference rooms should require more attention, where soundproof materials can result in employees who have not evacuated according to the responders' command.
- **Industrial Facilities:** In factories, the primary principle is to isolate areas with hazardous materials. The model ensures that these locations receive a mandatory second check, but the more attention paid the better.

We are confident that integrating this framework into our community's currently existing emergency protocols will lead to more rapid, safer evacuations. Our team is well prepared to provided continued support for the Committee in the initial implementation of this strategy, including the construction of building maps and simulation for key facilities.

We appreciate your consideration for this critical matter and look forward to collaborating for our community.

Best regards,

Team #16471

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## Report on the Use of AI

DeepSeek (November 13, DeepSeek-V3.2)

Query: Give me some real scientific articles on priority of certain areas during evacuation.

Output: See [2].

DeepSeek (November 14, DeepSeek-V3.2)

Query: Improve the grammar of this passage, but keep original content, diction, and sentence structure.

Output: The introduction paragraph and the conclusion paragraph. (Validated and further edited by team members, the AI's result is just for identifying grammar mistakes.)

DeepSeek (November 14, DeepSeek-V3.2)

Query: Rodrigo Paredes and Gonzalo Navarro. 2006. Optimal incremental sorting. In *Proceedings of the Meeting on Algorithm Engineering & Experiments*. Society for Industrial and Applied Mathematics, USA, 171–182. Change this ACM ref style to APA.

Output: Paredes, R., & Navarro, G. (2006). Optimal incremental sorting. *In Proceedings of the Meeting on Algorithm Engineering and Experiments* (pp. 171–182). Society for Industrial and Applied Mathematics.